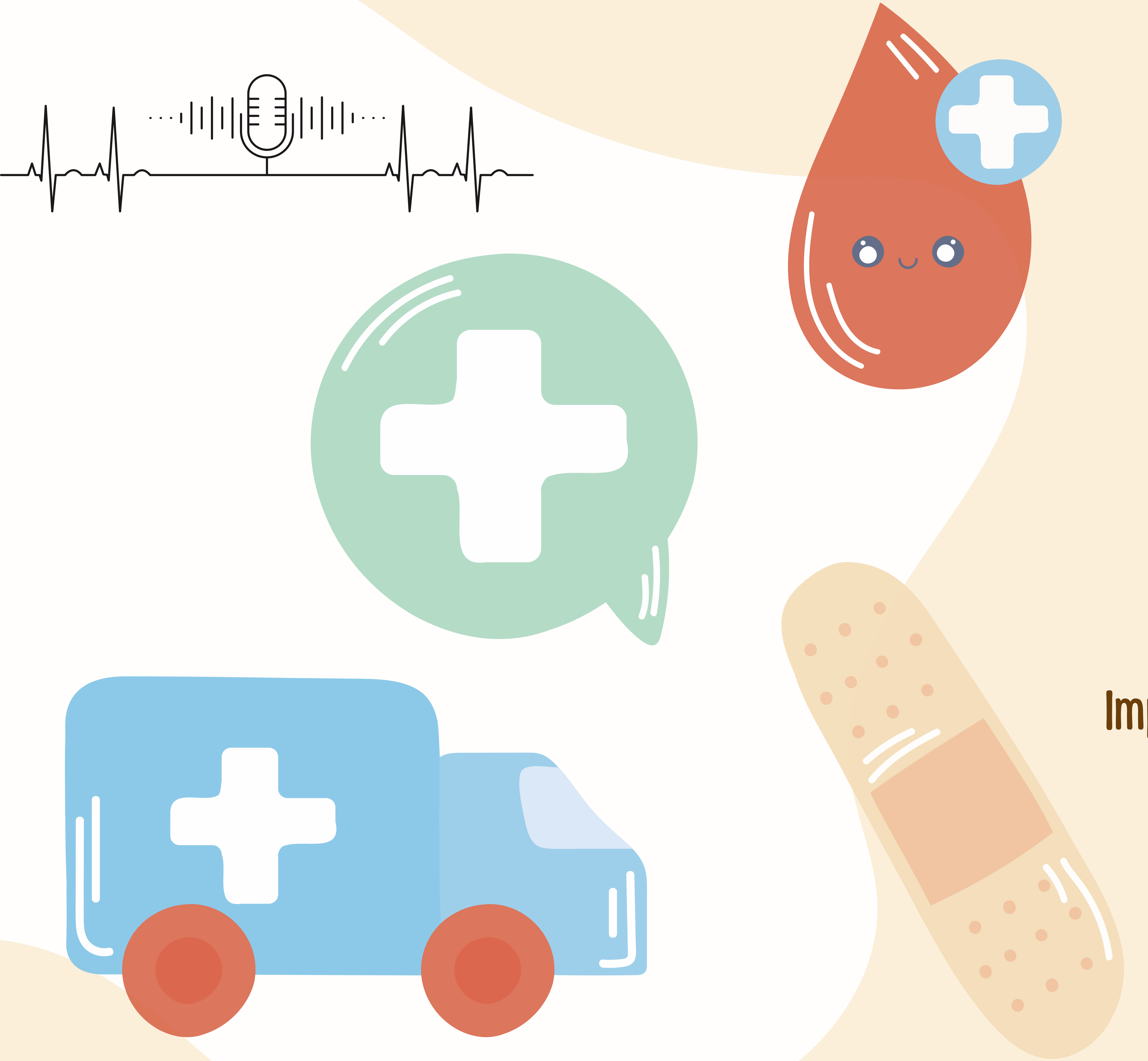


MedAlert

ED DECISION CLASSIFICATION FROM LLM-GENERATED EMS VOICE REPORTS

Improving Robustness to ASR Noise with Clean and Noisy Training

STUDENTS:

Nofar Refaeli ,Meital Gerasi

PROJECT DEFINITION

MOTIVATING USE CASE

EMS pre-arrival reports provide early clinical information that may help the ED prepare the appropriate care area and specialty consultation.

However, these reports are often brief, incomplete, and communicated verbally under noisy ambulance conditions.

To simulate this setting, we converted the reports to speech, added ambulance noise, and transcribed them using ASR.

PROBLEM DEFINITION

Task:

Predict ED preparation decisions from EMS pre-arrival handoff reports.

input:

Synthetic EMS pre-arrival handoff report

Outputs:

1. ED care area (arrival_destination)

2. Primary specialty consultation (specialty_consult_label)

Main research question:

How well can ED decisions be predicted from EMS reports, and does noisy ASR training improve performance?

Models:

DistilBERT and BioClinicalBERT

NOVELTY

We generate synthetic noisy EMS training data and train one Transformer model directly on the full report.

WE DO NOT USE A TRADITIONAL ASR → INFORMATION EXTRACTION →
RULE-BASED DECISION PIPELINE.

The model learns from both the clinical context and recurring ASR errors.
We test whether noisy and clean+noisy training improve downstream ED decision prediction.

DATASET PREPROCESSING

• **Source:** MIMIC-IV-Ext CDS task-specific files merged by stay_id.
• **Files used:** triage_level.csv, diagnosis.csv, and specialty_referral.csv.
• **Working subset:** 2,139 complete cases with HPI, initial vitals, triage, diagnosis, and specialty referral.

Cleaned noisy diagnosis/specialty fields:
specialties 591 → 63,
diagnoses 1,886 → 1,797

Parsed initial vitals from text into numeric variables and removed clinically implausible values.

Labeling

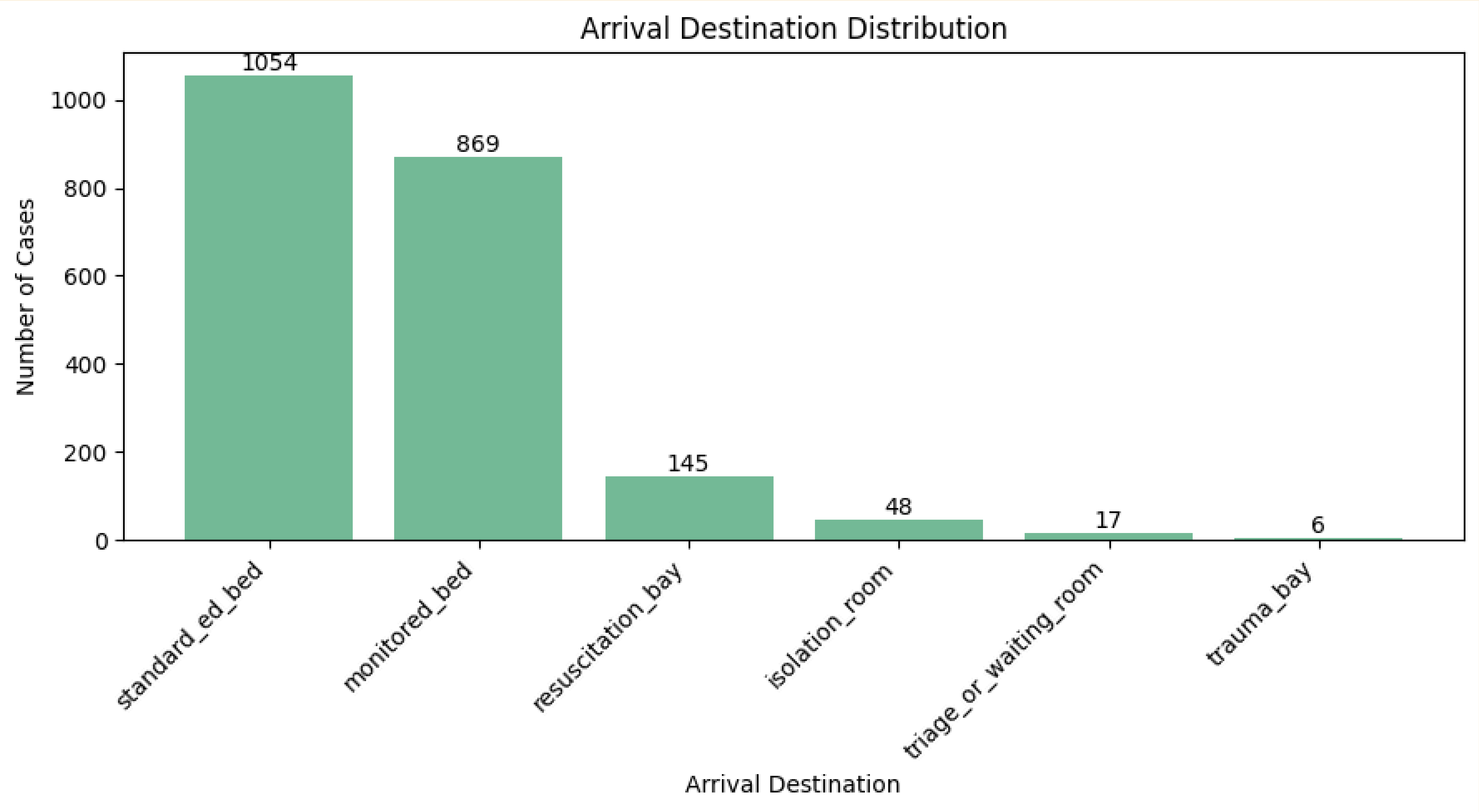
FIRST LABEL: ED CARE AREA

- Created using rule-based weak labeling.
- Input signals for labeling: clinical narrative, triage level, and parsed vital signs.
- Rules identify clinical instability, isolation risk, trauma/high-risk injury patterns, and low-acuity presentations.
- Used as target only; excluded from the text-generation prompt.

6 Final classes:

standard_ed_bed, monitored_bed, resuscitation_bay, isolation_room, trauma_bay, triage_or_waiting_room

TARGET LABEL DISTRIBUTION AND CLASS IMBALANCE



Labeling

SECOND LABEL: PRIMARY SPECIALTY CONSULTATION

• Derived from the cleaned specialty field.

• Original specialties were grouped into broader clinical consult labels.

• 63 original specialties were reduced to **18 final consult classes.**

• Generic labels such as medicine_or_ed_consult and other_consult were used only when no specific consult label was available.

• Used as target only; excluded from the text-generation prompt.

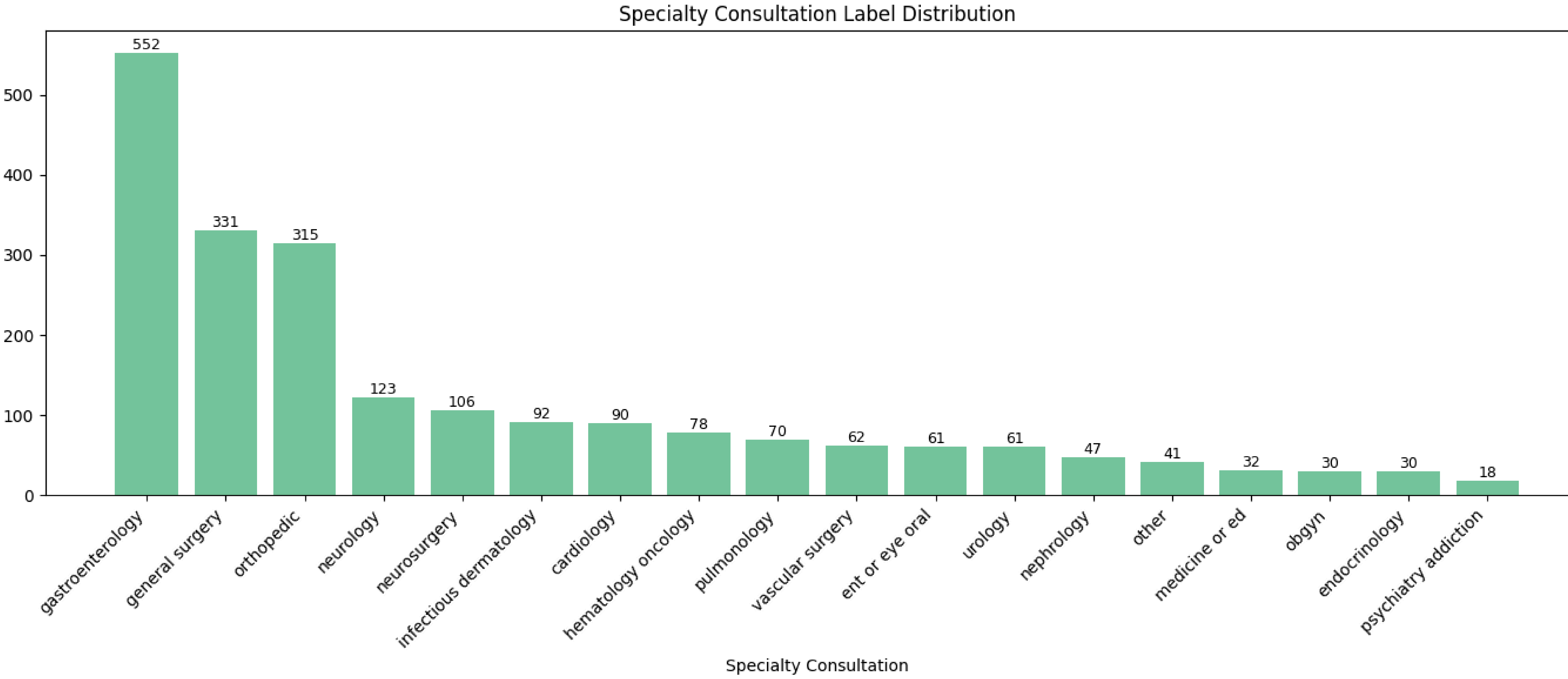
Examples:

orthopedics / hand surgery / podiatry → orthopedic_consult

ENT / ophthalmology / oral surgery / dentistry → ent_or_eye_oral_consult

**We intentionally prioritize ASR robustness over multi-label modeling in order to keep the experimental question focused and interpretable.*

TARGET LABEL DISTRIBUTION AND CLASS IMBALANCE



LEAKAGE-AWARE CASE SEED FOR REPORT GENERATION

- Each case was converted into a structured seed containing HPI, patient information, and initial vital signs.
- To reduce leakage, the seed excluded target labels, specialty fields, diagnosis fields, and triage, since these were related to the target-label creation.


```
{
  "source_case_id": 32822973,
  "patient_info": "Gender: Female, Race: WHITE, Age: 80",
  "hpi": "Patient is an ___ year-old patient with history of Sjogren's\nsyndrome, moderate MR, recent hospi
  "initial_vitals_raw": "Temperature: 100.2, Heartrate: 95.0, resprate: 28.0, o2sat: 100.0, sbp: 99.0, dbp:
  "vitals": {
    "temperature": 100.2,
    "heart_rate": 95.0,
    "respiration_rate": 28.0,
    "oxygen_saturation": 100.0,
    "systolic_bp": 99.0,
    "diastolic_bp": 47.0
  }
}
```

SYNTHETIC EMS REPORT GENERATION

2,139 LABELED CLINICAL CASES

↓

LEAKAGE-AWARE CASE SEEDS

↓

GPT-4.1-MINI TEXT GENERATION

↓

4 EMS REPORT VARIANTS PER CASE

↓

8,556 SYNTHETIC EMS REPORTS

4 VARIANTS PER CASE:

1. professional_complete
2. brief_radio_missing_details
3. patient_reported_uncertain
4. distracted_or_disorganized_handoff

FINAL DATASET LABEL DISTRIBUTION

**The generated dataset preserves the original class imbalance, since each source case produced four report variants.*

```
Specialty consult label distribution:
specialty_consult_label
gastroenterology_consult           2208
general_surgery_consult            1324
orthopedic_consult                 1260
neurology_consult                   492
neurosurgery_consult                424
infectious_dermatology_consult      368
cardiology_consult                  360
hematology_oncology_consult         312
pulmonology_consult                 280
vascular_surgery_consult            248
ent_or_eye_oral_consult             244
urology_consult                     244
nephrology_consult                  188
other_consult                       164
medicine_or_ed_consult              128
obgyn_consult                       120
endocrinology_consult               120
psychiatry_addiction_consult         72
```

```
Arrival destination distribution:
arrival_destination
standard_ed_bed            4216
monitored_bed              3476
resuscitation_bay           580
isolation_room              192
triage_or_waiting_room       68
trauma_bay                   24
```

Example: Same Clinical Case, Different EMS Report Variants

```
Source case ID: 30001856
Arrival destination: standard_ed_bed
Specialty consult label: gastroenterology_consult

-----------------------------------------------------------------------------------------
Text variant: professional_complete
-----------------------------------------------------------------------------------------
We are en route with a 54-year-old male with a history of alcoholic cirrhosis and hepatitis C,
status post TIPS procedure, presenting with three weeks of worsening epigastric pain now
constant and radiating around his abdomen. He reports vomiting this morning, possibly with
blood, and subjective fever and chills earlier today. No chest pain or shortness of breath.
Vital signs are stable with a temperature of 99.1, heart rate 83, respiratory rate 16, oxygen
saturation 97%, and blood pressure 140/71. Patient remains awake and alert.

-----------------------------------------------------------------------------------------
Text variant: brief_radio_missing_details
-----------------------------------------------------------------------------------------
54-year-old male with history of cirrhosis and hepatic encephalopathy, presenting with
persistent epigastric pain radiating around the abdomen. Vomited this AM, possible blood noted.
Vitals stable, BP 140/70, O2 sat 97%, no respiratory distress. Transporting alert, pain
ongoing.
```

POST-PROCESSING AND LEAKAGE REDUCTION

After generation, reports were scanned for hospital-based or leakage-prone wording.

DETECTED PATTERNS

- Imaging/lab results
- Specialist consult wording
- Hospital workflow phrases

METHOD

Regex-based candidate detection → minimal LLM rewrite → second-pass check.

RESULTS

- Round 1 candidates: 461 reports (5.39%)
- Round 2 remaining candidates: 168 reports (1.96%)

POST-PROCESSING AND LEAKAGE REDUCTION

The goal was to preserve clinical meaning while keeping the text EMS-style.

```
--- ORIGINAL TEXT ---
51-year-old male with history of esophageal cancer post-esophagectomy, presenting with 2
days of worsening crampy abdominal pain and bilious vomiting. CT shows small bowel
obstruction. Vitals stable with BP 102 systolic, O2 sat 95% on room air. Transporting now.

--- CLEANED TEXT ---
51-year-old male with history of esophageal cancer post-esophagectomy, presenting with 2
days of worsening crampy abdominal pain and bilious vomiting consistent with small bowel
obstruction. Vitals stable with BP 102 systolic, O2 sat 95% on room air. Transporting now.
```

TRAIN / VALIDATION / TEST SPLIT

Split strategy:

- Group-based split by source_case_id: all 4 variants of the same clinical case were kept in the same split
- Class coverage check: all target classes are represented in train, validation, and test.

This prevents leakage between train, validation, and test.* ***The same group-based split was preserved after ASR generation.*

Final split:

70% TRAIN:

5,988 reports (1,497 source cases)

15% VALIDATION:

1,284 reports (321 source cases)

15% TEST:

1,284 reports (321 source cases)

ASR ROBUSTNESS PIPELINE

Example of audio after adding noise

GOAL

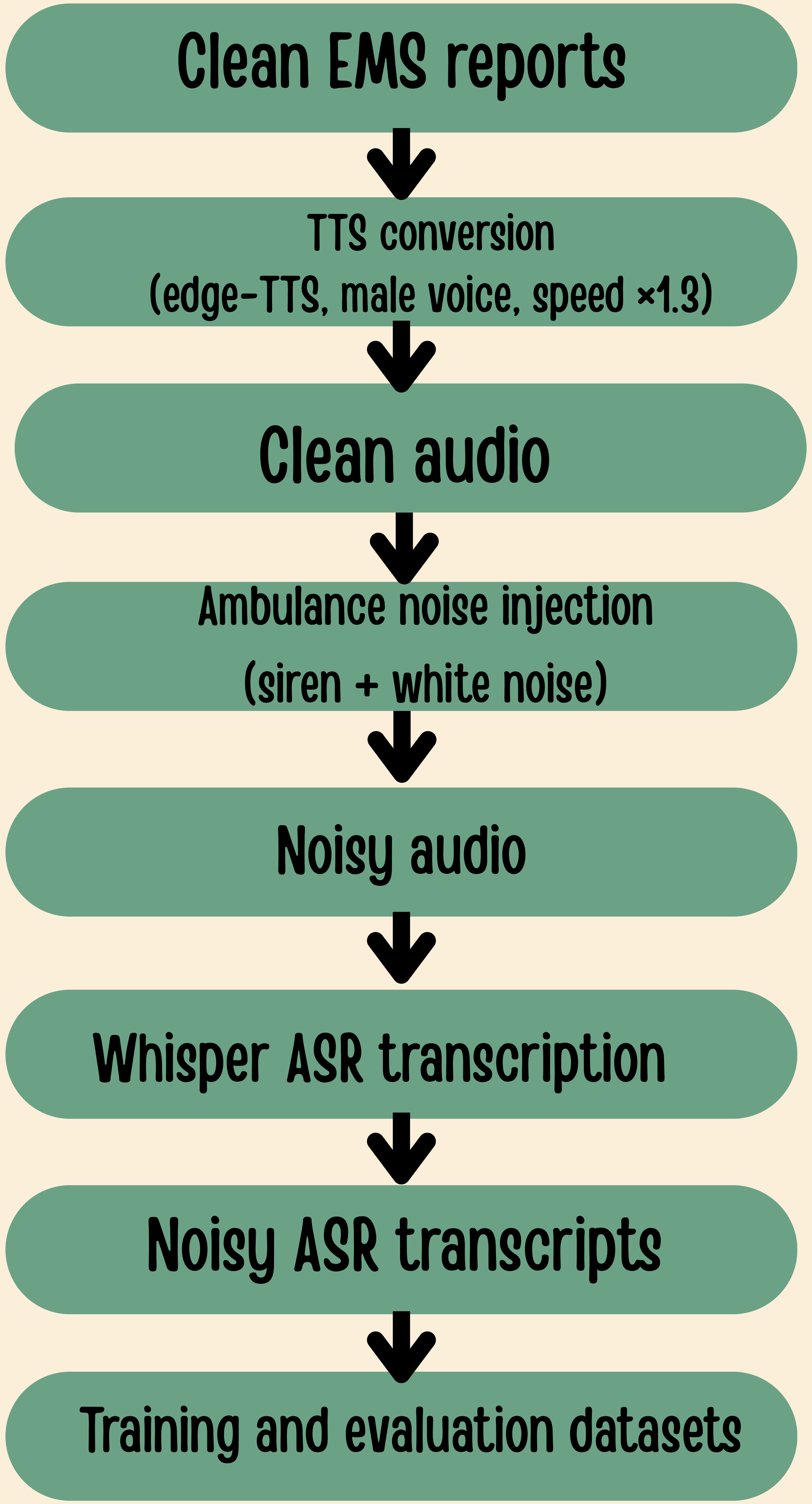
Test whether training with noisy ASR transcripts improves model robustness.

Training strategies:

- Clean text only
- Noisy ASR text only
- Clean + noisy ASR text

Evaluation:

The original source-case split was preserved, and the same noisy test set was used for all comparisons.



ASR Error Example: Clean Report vs Noisy Transcript

Mean WER: 0.443 ≈ 44%

WER measures the word-level difference between the clean EMS report and the ASR transcript.

noise dataset:
Average WER:
Train: 44.3%
Validation: 43.9%
Test: 44.8%

Original cleaned text:
We are transporting a 46-year-old female with a history of Crohn's disease who is experiencing sudden onset upper abdominal cramping radiating to her chest and back, consistent with her past Crohn's flares. She reports nausea and four episodes of bilious vomiting with small blood specks but denies diarrhea or constipation. Her vitals are stable: heart rate 72, blood pressure 132/65, respiratory rate 16, oxygen saturation 100%, and temperature 97.4. She has not taken her Humira injection in over two weeks. Patient is currently stable.

ASR noisy transcript:
We are transporting a 46-year-old female with a history of proven disease who was experiencing sudden onset of her abdominal cramping radiant or chest and back, consistent with her past crows' flares. She reports nausea and four episodes of billis fondly with small blood spests of deni-ciaria or constipation. Her vitals are stable, heart rate 72, blood pressure 152-165, risk occurring rate 16, oxygen saturation 100%, and temperature 97.4. She has not taken her cumulative injection in over two weeks. Patient is currently stable.

MODELS AND TRAINING STRATEGIES

DistilBERT:

- General-purpose language model
- Smaller and faster baseline
- Included to test whether a general model
maybe less sensitive to ASR errors

BioClinicalBERT:

- Pre-trained on biomedical and clinical text
- Better suited to medical terminology
- Used as the main domain-specific model

EXPERIMENTAL STAGES

Stage 1 – Clean Baseline

- Train on clean text
- Test on clean text
- Establish the best baseline model
- for each target

Stage 2 – ASR Robustness Evaluation

- Train clean → Test noisy
- Train noisy → Test noisy

**Clean + noisy training → Noisy test (BioClinicalBERT only)*

BEST BASELINE MODELS ON CLEAN TEXT

Optimization experiments

DistilBERT vs. BioClinicalBERT | Full fine-tuning vs. partial freezing | Weighted vs. unweighted loss | 3 vs. 5 epochs | Learning rate: 2e-5 vs. 1e-5 | Max length: 128 vs. 192

1. ED Care Area

Best model: BioClinicalBERT

- Partial fine-tuning: last 8 layers trainable
- Max length: 128
- 3 epochs
- Learning rate: 2e-5
- No class weights

Clean test results

Accuracy: 67.1%

Macro F1: 0.319

Weighted F1: 0.652

2.Primary Specialty Consultation

Best model: BioClinicalBERT

- Full fine-tuning
- Max length: 192
- 3 epochs
- Learning rate: 2e-5
- No class weights

Clean test results

Accuracy: 65.7%

Macro F1: 0.506

Weighted F1: 0.640

BEST DISTILBERT BASELINE MODELS ON CLEAN TEXT

Selected configuration

Full fine-tuning | No class weights | 3 epochs | Learning rate: 2e-5 | Max length: 128

ED Care Area

Clean test results:

Accuracy: 66.5%

Macro F1: 28.0%

Weighted F1: 63.8%

Primary Specialty Consultation

Clean test results:

Accuracy: 60.7%

Macro F1: 34.4%

Weighted F1: 57.6%

1. CLEAN TRAINING → NOISY TEST

Model	Target	Clean Test Macro F1	Noisy Test Macro F1	Relative Change in Macro F1 (%)
DistilBERT	ED care area	28.00%	23.87%	−14.75%
BioClinicalBERT		31.86%	22.55%	−29.22%
DistilBERT	Specialty consultation	34.40%	22.45%	−34.74%
BioClinicalBERT		50.60%	27.74%	−45.18%

1. CLEAN TRAINING → NOISY TEST

Model	Target	Clean Test Accuracy	Noisy Test Accuracy	Change
DistilBERT	ED care area	66.50%	55.90%	−10.6 pp
BioClinicalBERT		67.10%	59.00%	−8.0 pp
DistilBERT	Specialty consultation	60.70%	40.80%	−19.9 pp
BioClinicalBERT		65.70%	39.30%	−26.4 pp

2. TRAIN NOISY ASR → TEST NOISY ASR

Model	Target	Clean → Noisy Macro F1	Noisy → Noisy Macro F1	Relative Change in Macro F1
BioClinicalBERT	ED care area	22.55%	**27.64%**	22.57%+
DistilBERT		23.87%	21.98%	-7.92%
BioClinicalBERT	Specialty consultation	27.74%	**31.21%**	12.51%+
DistilBERT		22.45%	19.82%	-11.71%

Noisy training improved BioClinicalBERT in both targets, but did not improve DistilBERT according to Macro F1.

2. TRAIN NOISY ASR → TEST NOISY ASR

Model	Target	Clean → Noisy Accuracy	Noisy → Noisy Accuracy	Change
BioClinicalBERT	ED care area	59.03%	**64.64%**	+5.61 pp
DistilBERT		55.92%	62.46%	+6.54 pp
BioClinicalBERT	Specialty consultation	39.25%	**49.77%**	+10.52 pp
DistilBERT		40.81%	45.09%	+4.28 pp

EXAMPLE: NOISY TRAINING IMPROVED ROBUSTNESS

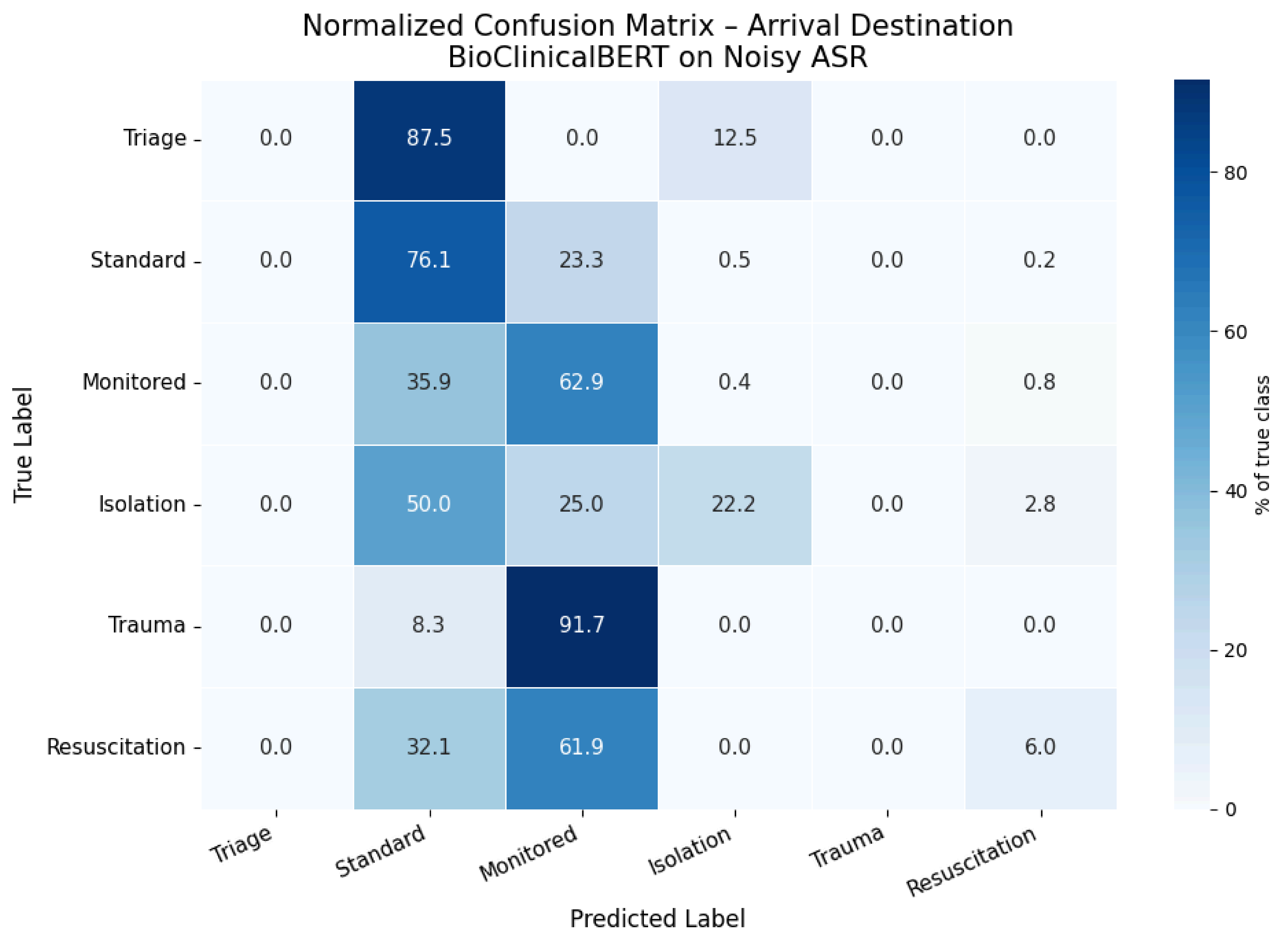
```
================================================================================
EXAMPLE: NOISY TRAINING CORRECTED THE PREDICTION
================================================================================

True specialty: orthopedic_consult
Clean-trained model prediction: ent_or_eye_oral_consult
Noisy-trained model prediction: orthopedic_consult
WER: 64.91%

Clean report
------------
We’re bringing in a 27-year-old male who slipped on stairs carrying laundry and fractured
his left distal tibia. He denies passing out or hitting his head, no other pain, numbness,
or tingling reported. He’s holding the ankle due to pain but no movement. Vitals are
stable: pulse 80, breathing 16, BP 140/70, oxygen 98% on room air.

noise ASR transcript
--------------------
We're bringing in a 27-year-old nail whose lip comes there to carry one re-intruch of his
leftus with tidius. He denies passing out or hitting his head, no other being, none as
once in the world reported. He's holding the ankle through the head, but no moving. Witals
are stable, plus 80, breathing 16, if he wants 40, slightly 70, oxygen 98% of the room
air.
```

3. CLEAN + NOISY TRAINING → NOISY TEST: BioClinicalBERT

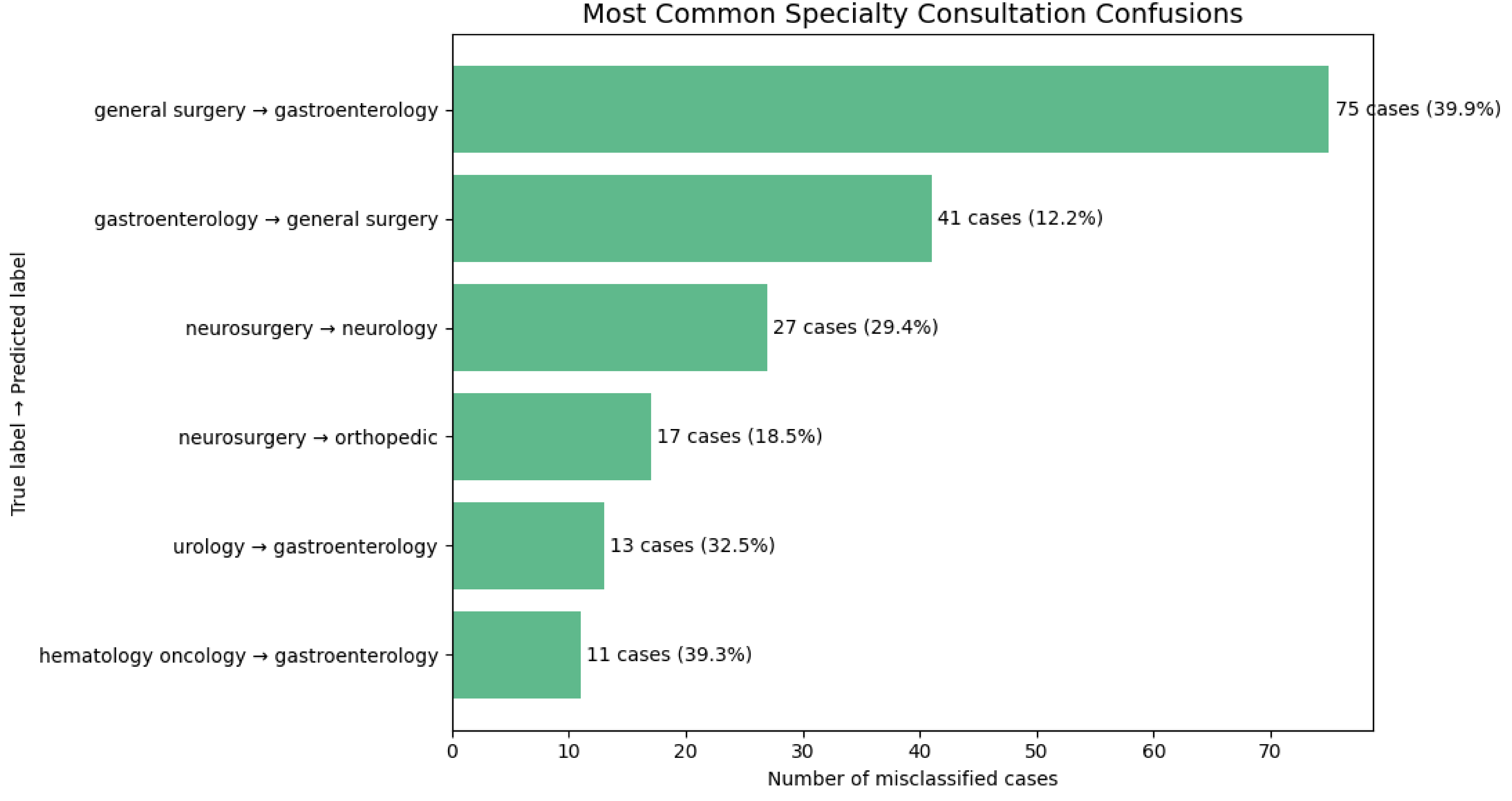
Target	Clean → Noisy Macro F1	Noisy → Noisy Macro F1	Clean + Noisy → Noisy Macro F1
ED care area	22.55%	27.64%	**29.16%**
Specialty consultation	27.74%	31.21%	**38.33%**

Target	Clean → Noisy Accuracy	Noisy → Noisy Accuracy	Clean + Noisy → Noisy: Accuracy
ED care area	59.03%	64.64%	**63.63%**
Specialty consultation	39.25%	49.77%	**52.50%**

Error Analysis



Error Analysis



Error Analysis

11.48% of the apparent errors were valid secondary specialties

```
================================================================================
VALID SECONDARY PREDICTION IN THE DATASET ANNOTATION
================================================================================

Case summary:
48-year-old male with diabetes, spreading gangrene and infection in the right
foot, with gas in the tissues and need for urgent surgical intervention.

Primary true label:
General Surgery

Model prediction:
Vascular Surgery

Valid specialty annotations:
General Surgery | Orthopedic | Vascular Surgery

Why is this important?
The prediction was counted as an error in the single-label evaluation, although
Vascular Surgery was marked as a valid consultation for this case.
================================================================================
```

CONCLUSIONS AND FUTURE WORK

The solution improved robustness, but did not fully solve rare-class errors.

Performance was limited by ASR noise, class imbalance, label overlap, and partial multi-label ambiguity.

Oversampling and class-balanced training

More rare-class data

Multi-label specialty classification

Focus first on specialty consultation or ED care area, then extend to the second task.

Optimize the clean-to-noisy training ratio.